MiniSWG (Mini Secure Web Gateway)

Asynchronous Forward Proxy, Trie Policy Engine & Admin Console

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Repository: github.com/aneeshjaladi/MiniSWG
Architecture: Zero-Trust Edge / ZIA Inspired

1. REPOSITORY & DEPLOYMENT

GitHub Repository: https://github.com/aneeshjaladi/MiniSWG

Live Demo / Deployed Link: Operates locally as a desktop gateway proxy on `http://127.0.0.1:8080`. It integrates directly with client operating systems via automated batch scripts and an embedded web admin console at `http://127.0.0.1:8080/admin`.

2. TECHNOLOGY STACK

- **Backend:** Python 3.11+ / 3.13 built entirely on native `asyncio`. Handles cooperative non-blocking network I/O, raw socket streaming, and HTTP protocol parsing from scratch without third-party web frameworks (using standard library `asyncio`, `socket`, `urllib.parse`, `json`, and `winreg`).
- **UI:** Embedded Single-Page Application (SPA) served directly from the MiniSWG proxy. Engineered with Vanilla HTML5, responsive CSS3 (dark-mode theme, glassmorphism, Google Fonts Outfit/Inter), and native JavaScript `fetch` with live polling.
- **Storage & Persistence:**
  - **Logs:** High-fidelity JSON Lines (JSONL) append-only telemetry stream (`logs/access.log`).
  - **Policies:** JSON configuration (`config/policy.json`) compiled into an in-memory reverse-label hierarchical Domain Trie.
  - **Database:** Zero-dependency file storage + in-memory state for minimal footprint and microsecond lookups.

3. WORKING FEATURES (IMPLEMENTED & VALIDATED)

- ✓ **HTTP Forwarding:** Full parsing of request line and headers, validation, origin connection, and streaming bidirectional response forwarding.
- ✓ **HTTPS via CONNECT Tunnel (RFC 9110):** Blind bidirectional TCP byte streaming via `asyncio.gather` with strict domain-level policy gating prior to tunnel establishment.
- ✓ **Trie-Based Wildcard Domain Blocking:** Reverse-label hierarchical Trie achieving $O(L)$ matching, supporting exact hostnames and wildcard subdomains (e.g. `*.casino.com`, `*.badsite.com`).
- ✓ **Category Blocking:** Maps domains to threat and corporate categories (e.g. Malware, Gambling, Social Media) and blocks requests according to active policy rules.
- ✓ **Per-Client Rate Limiting (Token Bucket):** Client-IP rate limiting using a lazy-refill monotonic clock algorithm (`time.monotonic()`) with burst allowance; returns `429 Too Many Requests` with `Retry-After`.
- ✓ **Structured Telemetry Logging:** Detailed JSON log entries tracking timestamp, client IP, method, host, port, verdict action (`allow`, `block`, `rate_limited`), reason, category, status code, byte metrics, and latency in milliseconds.
- ✓ **Zero-Downtime Policy Hot-Reload:** Live policy updates without dropping active connections via asynchronous file watcher, POSIX `SIGHUP` handler, and web admin API triggers.
- ✓ **Custom Block Page:** Branded HTML block page returned with HTTP `403 Forbidden` for blocked web requests, and clean raw 403 termination for HTTPS CONNECT requests.
- ✓ **Windows System Proxy Integration:** Seamless registry automation (`winproxy.py`) to dynamically route and restore system-wide browser traffic (Edge, Chrome, Brave) to MiniSWG with 1 click.
- ✓ **1-Click Launchers:** `Start_MiniSWG.bat` and `Stop_MiniSWG.bat` providing immediate execution and clean restoration for end-users.
- ✓ **Integrated Benchmark Harness:** Built-in multi-concurrency benchmarking suite (`bench/run_bench.py`) measuring throughput and tail latencies.

** Note: Unimplemented enterprise features (such as JWT user authentication or SSL/TLS MITM decryption) have been omitted.*

4. WEB ADMIN DASHBOARD CAPABILITIES

Accessible at `http://127.0.0.1:8080/admin`, the dashboard enables non-technical and security admins to:

- **Live Telemetry & Activity Feed:** Auto-refreshing feed displaying client IP, request method, destination host, action badge (allow / block / rate_limited), category, status, and latency.
- **System Counters:** Instant visibility into uptime, total requests processed, and threat blocks enforced.
- **1-Click Windows Proxy Switch:** Global header control to route all workstation traffic through the gateway or restore direct connection.
- **Visual Category Policy Management:** One-click toggles to instantly enable or disable blocking for Malware, Gambling, and Social Media.
- **Custom Rules & Allowlist Creator:** Web form to add new domain rules or allowlist entries without manual JSON editing.
- **Interactive Domain Inspector:** Real-time domain tester returning immediate policy verdicts, matched rules, and contextual syntax hints.
- **Rate Limit & Policy Editor:** Configuration panel to adjust requests-per-minute and burst thresholds with instant hot-reload.

5. MEASURED BENCHMARK RESULTS

MiniSWG was benchmarked across two distinct architectures: (1) **Production In-Cluster Kubernetes** using Google's `hey` load generator against an `nginx:alpine` origin, and (2) **Bare-Metal Local AsyncIO** socket pooling:

A. In-Cluster Kubernetes Benchmarks (Minikube + Google hey)

Configuration	Concurrency	Total Requests	Throughput	Avg Latency	p50 Latency	p95 Latency	p99 Latency	Success Rate
Direct NGINX (Baseline)	20 concurrent	1,000	43,931.7 req/s	0.4 ms	0.2 ms	1.2 ms	1.8 ms	100% (1,000/1,000)
MiniSWG Gateway	20 concurrent	1,000	**1,251.5 req/s**	15.9 ms	10.4 ms	48.9 ms	51.1 ms	100% (1,000/1,000)
MiniSWG Gateway	50 concurrent	5,000	**1,089.8 req/s**	45.7 ms	29.4 ms	77.8 ms	90.4 ms	100% (5,000/5,000)
MiniSWG Gateway	100 concurrent	5,000	**1,162.5 req/s**	85.6 ms	91.0 ms	109.8 ms	154.2 ms	100% (5,000/5,000)
MiniSWG Gateway	200 concurrent	5,000	**1,228.6 req/s**	161.3 ms	165.2 ms	201.5 ms	240.9 ms	100% (5,000/5,000)

B. Bare-Metal Local AsyncIO Benchmarks

Mode	Concurrency	Throughput (req/s)	Avg Latency	p50 Latency	p95 Latency	p99 Latency
Policy Enabled	10 concurrent	**1,563.5 req/s**	6.34 ms	6.35 ms	6.96 ms	7.45 ms
Policy Enabled	50 concurrent	**1,595.3 req/s**	29.05 ms	31.47 ms	33.04 ms	33.21 ms
Policy Enabled	200 concurrent	**1,766.2 req/s**	77.69 ms	87.88 ms	96.65 ms	97.44 ms
Policy Disabled	10 concurrent	1,610.2 req/s	6.16 ms	6.13 ms	6.65 ms	6.99 ms
Policy Disabled	50 concurrent	1,628.0 req/s	28.37 ms	31.55 ms	32.60 ms	32.75 ms
Policy Disabled	200 concurrent	1,721.6 req/s	77.89 ms	88.12 ms	95.80 ms	96.15 ms

Key Performance Insights:

- **100% Zero-Drop Reliability:** 16,000 total requests evaluated in Kubernetes with 0 failed, timed-out, or dropped connections (0.00% error rate).
- **Sub-Millisecond Policy Overhead:** Trie evaluation adds less than **0.18 ms** to average latency compared to uninspected traffic, proving that reverse-label Trie lookups are virtually zero-cost.
- **Controlled Tail Latency:** Under 200 concurrent connections, p95 and p99 scale smoothly with queue depth without connection starvation.

6. SCALE NUMBERS & VERIFICATION

- **Domain Patterns Tested:** Validated against exact host rules (`evil.example`), wildcard prefixes (`*.badsite.com` , `*.casino.com` , `*.facebook.com`), multi-label deep subdomains (`a.b.c.badsite.com`), and allowlist overrides (`internal.company.com`).
- **Automated Test Coverage: 44 automated unit and integration tests** across 6 test modules (`test_admin.py` , `test_bench.py` , `test_parser.py` , `test_policy.py` , `test_ratelimit.py` , `test_stage9.py`) via `pytest` + `pytest-asyncio` with a **100% pass rate**.
- **Concurrent Stress Smoke Test:** Standalone 50-client concurrent request validation (`tests/smoke.py`) verifying socket cleanup and connection resilience.

7. TEAM STRUCTURE & CANDIDATE ROLE

Solo Project by Aneesh Jaladi (aneeshjaladi)

Solely architected, implemented, and benchmarked the entire system end-to-end: designed the asynchronous socket streaming engine, RFC-compliant HTTP/CONNECT parser, reverse-label prefix Trie, token-bucket rate limiter, Windows registry automation, web admin console, and benchmark harness.

8. COURSEWORK & SECURITY CONCEPTS

- **Relevant Academic Coursework:**
  - **Computer Networks:** TCP/IP socket programming, HTTP/1.1 request/response framing, RFC 9110 CONNECT tunneling, proxy forwarding semantics, and latency percentile analysis.
  - **Operating Systems:** Asynchronous cooperative event loops (`asyncio`), non-blocking socket I/O multiplexing (IOCP / epoll), OS signal handling (`SIGHUP`), and Windows registry IPC.
- **Applied Security Concepts:** Enterprise Secure Web Gateway (SWG) architecture inspired by Zscaler Internet Access (ZIA), zero-trust egress domain inspection, DoS mitigation via token-bucket rate limiting, and structured audit telemetry.
- **Links:** GitHub repository: https://github.com/aneeshjaladi/MiniSWG